

DFT Project: Water Splitting Reaction Free Energy Project Description

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Prerequisite: Basic Python programming skills (variables, loops, and simple plotting) are required.

If the upcoming pages look dense at first sight, don't worry, the detailed tasks are listed in Section 3 and should be used as the structural template when preparing your report. Section 2 provides some helpful information on how to get started with the recommended Python package PySCF.

Project Description

This project takes you through a step-by-step computational study of the water splitting reaction:



moving systematically from pure electronic energies to room-temperature Gibbs free energies.

Note: Throughout the project, use the following sign convention:

$$\Delta E_{\text{r}} = E_{\text{products}} - E_{\text{reactants}} = (2E_{\text{H}_2} + E_{\text{O}_2}) - 2E_{\text{H}_2\text{O}}. \quad (2)$$

With this convention, the water-splitting reaction energy is positive. E thereby denotes the electronic energy obtained from a Hartree Fock or DFT calculation. Analogously, the reaction free energy ΔG_{r} can be calculated from the molecular Gibbs free energies:

$$\Delta G_{\text{r}} = G_{\text{products}} - G_{\text{reactants}} = (2G_{\text{H}_2(\text{g})} + G_{\text{O}_2(\text{g})}) - 2G_{\text{H}_2\text{O}(\text{l})}. \quad (3)$$

The molecular Gibbs energies are obtained from electronic energies E and thermodynamic corrections, as explained in Section 1.6. The reference literature value for ΔG_{r} is

$$\Delta G_{\text{r}}^{298\text{ K}} \approx +4.92\text{ eV}.$$

Remember to use consistent units throughout the calculation.

You will learn how to construct molecules, optimise geometries, compute reaction energies, study basis set convergence, compare different electronic structure methods, add thermodynamic corrections, and analyse the accuracy of your final result.

1 Background and Instructions

1.1 Constructing Molecules

Define each molecule by listing atomic symbols and their Cartesian coordinates (x, y, z) in Å. Using PySCF, you will code these coordinates into Python and build a molecule.

The three molecules considered in this project are H_2 , O_2 , and H_2O . Use the following molecular settings throughout the project unless stated otherwise:

Molecule	Charge	spin in PySCF	Recommended reference
H ₂	0	0	RHF/RKS
H ₂ O	0	0	RHF/RKS
O ₂	0	2	UHF/UKS

Here, **spin** means $N_\alpha - N_\beta$. The value **spin=2** is important because molecular oxygen is a triplet in its ground state.

1.2 Geometry Optimisation

Relax each molecule to its lowest-energy arrangement using a basic method, Hartree–Fock (RHF for H₂ and H₂O, UHF for O₂) with the STO–3G basis set:

- Run an optimisation routine that adjusts atomic positions until energy changes are minimal.
- Measure the final bond lengths: H–H in H₂, O–O in O₂, and both O–H bonds in H₂O.
- For the H₂O molecule, additionally measure the bond angle between the H–O–H bonds.
- Compare these bond lengths and the H–O–H angle with known experimental values.

You will see how theory predicts realistic geometries and learn about basis sets.

1.3 Reaction Energy

Calculate the reaction energy ΔE_r of the water splitting reaction in Eq. 1. Report the electronic energy of each optimised structure first, then calculate the reaction energy using Eq. 2.

Report the reaction energy in Hartree and convert it to eV using

$$1 \text{ Hartree} \approx 27.2114 \text{ eV}.$$

1.4 Basis Set Convergence

Investigate how the size of the basis set affects the calculated water splitting reaction energy ΔE_r :

- Calculate ΔE_r with the following basis sets: STO–3G, 3–21G, 6–31G, cc–pVDZ, cc–pVTZ, cc–pVQZ, cc–pV5Z.
- Plot ΔE_r versus basis set to observe when the energy stops changing significantly.
- Use the geometries optimised at HF/STO–3G. Then keep these geometries fixed while changing only the basis set. This means that any change in ΔE_r comes mainly from the basis set, not from a change in geometry.
- As an additional quantitative comparison, take the cc–pVQZ result as the most accurate available reference within this study and compare the other basis sets with it. If cc–pV5Z can be completed reliably, include it in the discussion as an additional check.

1.5 Method Hierarchy (Jacob’s Ladder)

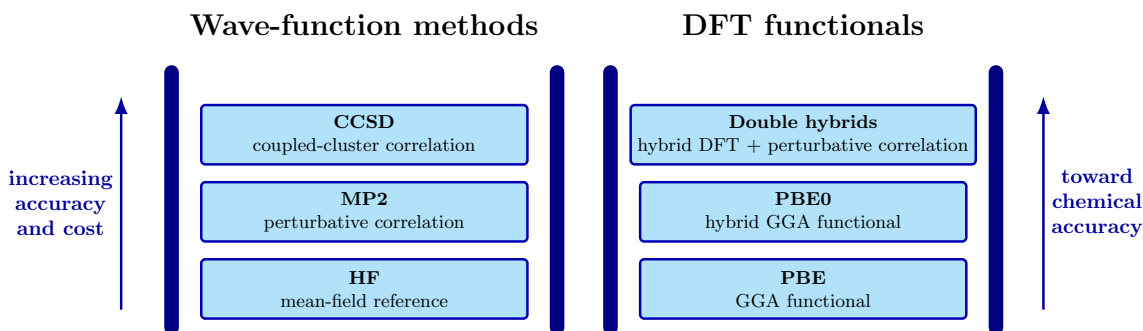


Figure 1: Schematic illustration of Jacob’s Ladder for wave-function and DFT methods. The rungs shown correspond to the methods and functionals considered in this project. Moving upward includes additional electronic structure information and is generally associated with improved accuracy and increased computational cost.

Fig. 1 presents Jacob’s Ladder as a conceptual framework for the hierarchy of approximations for wave-function and DFT methods, climbing from the simplest to the most accurate methods toward chemical accuracy. Improve upon Hartree–Fock by adding electronic correlation. You will:

- Calculate the reaction energy using HF, MP2, and CCSD for the water splitting reaction.
- In parallel, perform density functional theory (DFT) calculations: begin with the Perdew–Burke–Ernzerhof (PBE) approximation, which treats electron exchange with the Generalized Gradient Approximation (GGA). Proceed to a hybrid functional, for example PBE0, and discuss double hybrid options.
- With the basis set that has converged and that you decided to use, compute the reaction energy with the following method ladder:

$$\text{HF} \rightarrow \text{MP2} \rightarrow \text{CCSD}$$

and, in parallel,

$$\text{PBE} \rightarrow \text{PBE0 (hybrid)} \rightarrow (\text{double hybrid discussion})$$

If a high-level calculation becomes too expensive, use the largest basis set that can be completed reliably and discuss this limitation clearly.

1.6 Thermodynamic Corrections

When we compute the energy of a chemical system using DFT or a high-level wave-function method, we obtain the electronic energy at 0 K, without accounting for the motions of nuclei or the effects of temperature. To connect these raw energies to experiments carried out at room temperature, 298.15 K, we must add thermodynamic corrections that convert electronic energies into Gibbs free energies (G).

Main correction terms

- **Zero-point energy (ZPE):** Even at 0 K, each vibrational mode has a residual energy, $\frac{1}{2}\hbar\omega$. Adding ZPE accounts for the quantum mechanical ground state vibrations of bonds, for example the O–H bond stretches in water.
- **Thermal enthalpy correction (H):** At finite temperature, vibrations, rotations, and translations contribute extra thermal energy. Usually, H is computed from statistical mechanics formulas for each degree of freedom. In this project, you may use literature values for H . This raises the enthalpy from its 0 K value to the value at 298.15 K.

- **Entropy correction ($-TS$):** Molecules gain entropy through increased vibrational, rotational, and translational freedom at room temperature. Multiplying the entropy S by $-T$ gives the entropic contribution to the free energy.

Putting these together,

$$G(298.15\text{ K}) = E_{\text{DFT}} + \text{ZPE} + H - TS, \quad (4)$$

where E_{DFT} is the electronic energy from your quantum chemistry calculation.

Data source

- Look up experimental ZPE, H , and S values in the NIST database¹ and add them to your calculated electronic energies.
- (Optional) Compute vibrational frequencies in PySCF to estimate ZPE, H , and S yourself.

The Gibbs free energies of the respective molecules, H_2O , H_2 , and O_2 , can be used to compute the water splitting reaction energy in Eq. 1. If you have not calculated the vibrational frequencies yourself, adopt literature values for S and H . The literature values can be obtained from the NIST database. For gas-phase molecules such as H_2 and O_2 , compute G at 298.15 K according to Eq. 4.

An exception in the computation of G for molecules relevant to water splitting is the water molecule. Since water exists as a liquid at 298.15 K, its G is determined using the following expression:

$$G_{\text{H}_2\text{O(l)}}^{298.15\text{ K}} = G_{\text{H}_2\text{O(g)}}^{298.15\text{ K}} + (H_{\text{H}_2\text{O(l)}} - H_{\text{H}_2\text{O(g)}}) - T(S_{\text{H}_2\text{O(l)}}^{298.15\text{ K}} - S_{\text{H}_2\text{O(g)}}^{298.15\text{ K}}). \quad (5)$$

Important: Use consistent units throughout the calculation. Before adding electronic energies, ZPE, enthalpy corrections, and entropy terms, convert them to the same unit. Also remember that the reference value +4.92 eV corresponds to the reaction as written in Eq. 1.

1.7 Accuracy Analysis

Assess common approximations:

- Re-relax molecular geometries at higher levels, for example MP2 or CCSD, to include structural changes.
- Create a final accuracy analysis table summarising contributions from basis set, method, geometry, and thermodynamic corrections.
- Compare the obtained results with literature values.

2 Getting Started

In this project you will use the open-source quantum chemistry package **PySCF**². Make sure it is installed in your Python environment (`pip install pyscf` or `conda install -c conda-forge pyscf`). You can find the installation guide here <https://pyscf.org/install.html>. For the geometry optimization, you need to additionally install `geomeTRIC` (`pip install geometric` or `conda install -c conda-forge geometric`) or `PyBerny` (`pip install pyberny` or `conda install -c conda-forge pyberny`).

For Windows users: Currently, PySCF is not supported natively on Windows. To use it nonetheless, you need to install a Windows subsystem for Linux first. We recommend using WSL with Ubuntu as distribution. Instructions for the installation can be found here: <https://learn.microsoft.com/de-de/windows/wsl/install>. Usually, all that needs to be done is opening the Windows PowerShell as admin, and running the

¹<https://cccbdb.nist.gov/>

²<https://pyscf.org>

command `wsl --install`. After restarting the computer, a username and password needs to be set and you can set up your Python environment within the linux subsystem (e.g. by installing Miniforge/Mamba). If the shell for setting username and password does not open automatically, it can happen that no distribution was installed. In that case, after installation of WSL the distribution has to be downloaded separately, for the case of Ubuntu from the following page: <https://ubuntu.com/download/wsl>. After download, you can execute the file simply by opening it.

Minimal PySCF workflow

Pseudocode 1 Building and initialising a molecule

Require: Atom labels $\{\text{symbol}_i\}$ and coordinates $\{\mathbf{r}_i\}$ in Å

```
1: mol ← gto.Mole()
2: for all  $i$  do
3:   Append string "<symbol> x y z" to mol.atom
4: end for
5: Assign a basis
6: mol.unit ← "Angstrom"
7: MOL(build)
```

An SCF calculation means a self-consistent field calculation. For Hartree–Fock and DFT methods, the molecular orbitals and electron density are updated iteratively until the total electronic energy E is converged. The term SCF energy therefore refers to the converged electronic energy obtained from this iterative process.

PySCF helpful commands

Listing 1: PySCF code snippets

```
1 # load modules
2 from pyscf import gto, scf, mp, cc
3 from pyscf.geomopt.geometric_solver import optimize
4
5 # 1. Molecule container
6 mol = gto.Mole() # empty Mole object
7 mol_H2 = gto.M(atom="H 0 0 0; H 0 0 0.74", # example molecule
8             basis="sto-3g")
9
10 # 2. Mean-field calculation
11 mf = scf.UHF(mol).run() # unrestricted Hartree-Fock
12 print(mf.e_tot) # total energy (Eh)
13
14 # handy attributes
15 - mf.mo_coeff # molecular-orbital coefficients
16 - mf.converged # SCF convergence flag (true/false)
17
18 # 3. Geometry optimization
19 mol_opt = optimize(mf) # geometry optimization on the constructed molecule
20 mf_opt = scf.UHF(mol_opt).run() # UHF calculation on optimized molecule to obtain its energy
21
22 # handy attributes
23 - mol_opt.atom_coords() # molecular coordinates in bohr
24 - mf_opt.e_tot # energy of optimized molecule
25
26 # 4. Post-HF
27 e_mp2 = mp.MP2(mf).run().e_tot # MP2
28 e_ccsd = cc.CCSD(mf).run().e_tot # CCSD
29
30 # 5. Unit conversion
31 HARTREE2EV = 27.2114 # 1 Hartree ~ 27.2114 eV
32 # precise value:
```

```
33 # from scipy.constants.physical_constants['Hartree energy in eV'][0]
```

Assigning basis sets per atom

In PySCF you can pass a **dict** such as

```
mol.basis = {'H': 'cc-pVQZ', 'O': 'cc-pVTZ'}
```

to use different basis sets for different elements.

Geometry optimisation in PySCF

For geometry optimisation, use either the `geomeTRIC` or `PyBerny` optimiser through PySCF. For example, with `geomeTRIC`:

Listing 2: Geometry optimisation

```
1 from pyscf.geomopt.geometric_solver import optimize
2
3 mol_opt = optimize(mf)
```

With `mf` being the result of a mean-field calculation as constructed in 2. If this does not work, check that `geometric` or `pyberny` is installed in the same Python environment as `pyscf`.

3 Tasks (to be completed & reported)

This section is the main checklist for the project report.

- T1 Coordinates.** Enter experimental geometries for the three molecules. Cite your source.
- T2 Geometry relaxation.** Optimise the geometries using Hartree–Fock with the STO–3G basis set as described in Geometry Optimisation
- T3 Geometry analysis.** Report the H–H bond length in H_2 , the O–O bond length in O_2 , the O–H bond lengths in H_2O , and the H–O–H angle. Compare these values with literature.
- T4 Reaction energy.** Calculate the reaction energy of the water splitting reaction using the optimised structures.
- T5 SCF total energy.** Record the converged Hartree–Fock energy for each molecule before starting the basis set study.
- T6 Basis set convergence.** Using the Pseudocode 2, generate a plot of reaction energy versus different basis sets:

STO–3G, 3–21G, 6–31G, cc-pVDZ, cc-pVTZ, cc-pVQZ, cc-pV5Z.

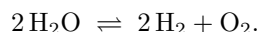
Also compare the other basis sets with the cc-pVQZ reference within this study to show how close each basis set is to the high-accuracy limit used in this project. If cc-pV5Z can be completed reliably, include it in the discussion as an additional check.

Pseudocode 2 Basis set convergence of the $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$ reaction energy

```
1: for basis  $\in \{\text{STO-3G}, 3\text{-}21\text{G}, 6\text{-}31\text{G}, \text{cc-pVDZ}, \text{cc-pVTZ}, \text{cc-pVQZ}, \text{cc-pV5Z}\}$  do      ▷ — build the
   three molecules in this basis —
2:   set_basis( $\text{H}_2$ , basis)
3:   set_basis( $\text{O}_2$ , basis)
4:   set_basis( $\text{H}_2\text{O}$ , basis)

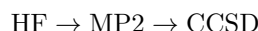
                                       ▷ — SCF energies in Hartree —
5:    $E_{\text{H}_2} \leftarrow \text{run\_scf}(\text{H}_2, \text{method})$ 
6:    $E_{\text{O}_2} \leftarrow \text{run\_scf}(\text{O}_2, \text{method})$ 
7:    $E_{\text{H}_2\text{O}} \leftarrow \text{run\_scf}(\text{H}_2\text{O}, \text{method})$ 
8:    $\Delta E_{\text{r}} \leftarrow (2E_{\text{H}_2} + E_{\text{O}_2}) - 2E_{\text{H}_2\text{O}}$       ▷ — products minus reactants —
9:    $\Delta E_{\text{r}}(\text{eV}) \leftarrow \Delta E_{\text{r}}(\text{Hartree}) \times 27.2114$ 
10:  Append (basis,  $\Delta E_{\text{r}}(\text{eV})$ ) to list
11: end for
```

T7 Single molecule vs reaction energy convergence. For the water molecule, plot the convergence of its total SCF energy with respect to basis set along with the convergence of the total reaction energy:

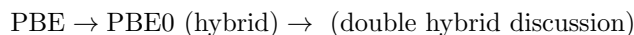


Discuss possible reasons why the reaction energy appears to converge faster than the total energy of the isolated molecule.

T8 Method ladder. With the basis set that converged and that you decided to use, compute the reaction energy with the following methods:



and, in parallel,



If a high-level calculation becomes too expensive, use the largest basis set that can be completed reliably and discuss this limitation clearly. Double hybrid calculations may also be discussed if they cannot be completed reliably.

T9 Zero-point energy and thermal corrections. Look up experimental ZPE, H , and S values in the NIST database³ and add them to your best electronic energies. Optionally, compute harmonic vibrational frequencies with PySCF and compare.

T10 Corrections and discussion.

- Examine how your reaction energy changes when you include (i) geometry relaxation at every new level of theory and (ii) the thermodynamic correction terms.
- Create a bar chart that displays, for each electronic structure method, the deviation of its thermodynamically corrected reaction energy from the literature value.
- Add a brief discussion that compares the methods and explains the sources of the observed errors.
- Use the literature value for the reaction free energy⁴

$$\Delta G_{\text{r}}^{298\text{ K}} \approx +4.92\text{ eV} \quad \text{for the water-splitting reaction.}$$

³<https://cccbdb.nist.gov/>

⁴Derived from the reverse of the Gibbs free energy of formation of liquid water, $\Delta G_f^\circ[\text{H}_2\text{O}(\text{l})] = -237.18\text{ kJ mol}^{-1}$ (NIST-JANAF Thermochemical Tables, 4th ed., 1998); <https://janaf.nist.gov>. Two moles of H_2O give $+474.36\text{ kJ mol}^{-1}$, which converts to $474.36/96.485 \approx +4.92\text{ eV}$, using $1\text{ eV} = 96.485\text{ kJ mol}^{-1}$.

Suggested report structure

The report can follow the task list above. A clear structure would be:

- Introduction and reaction definition.
- Computational setup.
- Geometry optimisation and geometry analysis.
- Reaction energy and basis set convergence.
- Method comparison.
- Thermodynamic corrections.
- Accuracy analysis and discussion.

Report guidelines

- Keep the final report approximately around 10 pages.
- State the sign convention and units clearly.
- Include snippets of your code wherever it supports your results.
- Discuss your numerical results and interpret what they mean in the context of electrocatalytic water splitting.

Hints & Resources

- **ZPE and thermodynamic data:** look up the CCCBDB at NIST for vibrational and thermochemical data.
- **O₂:** remember that O₂ is a triplet. Use an unrestricted reference with **spin=2**.
- **Units:** keep Hartree, eV, and kJ mol⁻¹ conversions consistent throughout the report.